# Titus Velpula

AI DEVELOPER

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https://measuredai.co.nz/portfolio/projects.html

## PROFILE

AI Developer with a data science background, building, testing, and deploying AI agents, automation workflows and API integrations for business operations. At Celtine Ltd, I work directly with operational teams and business owners to identify automation opportunities and deliver reliable solutions, combining business logic with AI and LLMs where they add value. Hands-on experience with JavaScript, SQL, Python and API integrations. I hold a Master of Analytics (Honours).

## KEY ACHIEVEMENTS

> **Agents in production:** Designed and deployed AI agents that classify enquiries, retrieve business data and generate contextual replies, with escalation to staff.

> **Reduced manual work:** Automated sales orders, supplier stock checks, out-of-stock handling and dropship tracking across multiple brands.

> **Concept to working system:** Built social-media automation connecting product data, AI content generation, media rendering, human review and scheduled publishing.

> **Evaluation and reliability:** Built output checks and evaluations to identify failures and compare retrieval accuracy, model performance and cost.

## TECHNICAL SKILLS

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| **Development and data** | JavaScript, Python, SQL, PostgreSQL, Supabase, Git |
| **AI engineering** | Claude and OpenAI APIs, prompt design, tool calling, structured outputs, context management, multi-agent orchestration, RAG, LangChain, Ollama, hybrid BM25 and dense retrieval |
| **Evaluation and observability** | Deterministic checks, LLM-as-judge, DeepEval, reference datasets, retrieval testing, human review, execution logging, failure detection, cost and latency monitoring |
| **Integration and infrastructure** | n8n, REST/GraphQL APIs, webhooks, authentication, Docker, Linux, Cloudflare Tunnel, credential and environment management, Claude Code, Claude Cowork |
| **Analytics** | Power BI, Tableau, Alteryx, Snowflake, MATLAB, SAS, R, statistics |

## WORK EXPERIENCE

### AI Developer · Celtine Ltd CONTRACT

May 2025 to present

- Collaborate with operational staff and business owners to map manual processes, identify the automatable core, and decompose work into steps an agent can own.
- Build AI agents and automation with n8n, JavaScript, APIs and webhooks, applying business rules for predictable processing and Claude/OpenAI models where language understanding and generation add value.
- Automate sales-order processing, supplier stock checks, out-of-stock notifications, dropship tracking and fraud screening by integrating e-commerce, ERP, helpdesk and shipping systems.
- Write evaluations and validate output; investigate production failures and eliminate silent failures; improve prompts, routing and error handling to strengthen reliability.

- Design for failure with graceful degradation and escalation to staff; keep prompts and instructions model-agnostic so providers can be swapped without a rewrite.
- Maintain the self-hosted Linux infrastructure supporting production automation (Docker, Cloudflare Tunnel, backups, credential management).

## PROJECTS

**RAG research assistant** · Personal project · github.com/TitusDV/rag-research-assistant 2026

- Built an assistant that answers questions about five AI research papers, cites paper and page for every claim, and says when the papers don't cover it (LangChain, OpenAI, Gradio).
- Compared BM25, dense and hybrid retrieval across four chunk sizes on 36 reviewed test questions (best MRR 0.79, hit@5 up to 0.97); traced dense retrieval's drop on long chunks to the embedding model's 256-token limit.
- Built an LLM judge for test-question review that agreed with my own review 94% of the time; tests run on every push with GitHub Actions.

**AI social media automation platform** · Celtine Ltd 2026 to present

- Built a multi-workflow agent system for content generation, media rendering and publishing across Instagram, Facebook, Threads and TikTok, using product and media data from Magento 2 and a Supabase/PostgreSQL layer.
- Integrate Claude, Creatomate and platform APIs with OAuth token handling, scheduled publishing, evaluation and human review; test media selection, rendering and publishing failures before production rollout.

**Telecare RAG chatbot** · Personal project 2024

- Built a document question-answering assistant in Python with LangChain, combining BM25 and sentence-embedding search, with a Gradio interface and a local Llama 3 model (Ollama) or OpenAI.

## ADDITIONAL WORK EXPERIENCE

**Telecare Monitoring Representative** · Hato Hone St John PART-TIME Oct 2023 to present

- Respond to emergency alarms and coordinate with customers, responders and support services; record incidents accurately and make timely escalation decisions under established protocols.

**Campaign Developer** · Digital Alchemy Oct 2021 to Feb 2022

- Built multi-channel marketing campaigns in Unica Campaign using customer segmentation; translated business briefs into campaign logic and completed pre- and post-campaign analysis with Oracle and SQL.

**Analytics Specialist** · Spark NZ (Skinny Mobile and Broadband) Jan 2021 to Jun 2021

- Built Power BI and Tableau dashboards and automated reporting with Alteryx, SQL and Snowflake; worked with product and marketing teams to turn analysis into practical recommendations.
- Awarded for best customer service.
- Provided thorough analysis and post-launch support for products and plans.

## EDUCATION AND CERTIFICATION

**Master of Analytics (Honours)** · Auckland University of Technology 2023 to 2024

- Studied statistics, machine learning and operations research. Thesis used support vector machines to classify stress from patients' brain EEG data, achieving 89% accuracy and 86.8% cross-validated accuracy, in collaboration with the New Zealand College of Chiropractic.

**SAS Certified Specialist: Base Programming Using SAS 9.4**

**Bachelor of Technology in Computer Science Engineering** · Jawaharlal Nehru Technological University, India 2007 to 2011

- Advanced OOP concepts, Database Management, SQL, Mathematics, Statistics.